RAJALAKSHMI ENGINEERING COLLEGE

M Maadhavan 250701394-2025-CSE

M2

Started on	Friday, 11 September 2026, 8:42 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:39 AM
Time taken	57 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<string.h>
int main(){
    char s1[1000], s2[1000];
    int dp[1000][1000];
    int i,j;
    scanf("%s", s1);
    scanf("%s", s2);
    int n=strlen(s1);
    int m=strlen(s2);
    for(i=0; i<=n; i++)
        dp[i][0]=0;
    for(j=0; j<=n; j++)
        dp[0][j]=0;
    for(i=1; i<=n; i++){
        for(j=1; j<=m; j++){
            if(s1[i-1]==s2[j-1])
               dp[i][j] = dp[i-1][j-1]+1;
            else
               dp[i][j] = dp[i-1][j]>dp[i][j-1]? dp[i-1][j] : dp[i][j-1];
        }
    }
    printf("%d\n", dp[n][m]);
    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course